

# Phase 1 (MVP) Subtask Deepening

## — epic → task → subtask (closes refinement R5)

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Volume 7 (Phase Execution), document 3 of 5. This spec **closes refinement item R5** (REFINEMENT\_NOTES.md): Phase 0 (06-...) formalizes a full epic→task→subtask 3-tier breakdown, but Phase 1 (07-phase1-mvp-wbs.md) stops at epic→task. Here every Phase-1 task HVPN-P1-NNN is decomposed into PR-sized subtasks HVPN-P1-NNN.k (§11.4.93/.54 ids), each carrying a concrete **acceptance criterion** (falsifiable, captured-evidence per §11.4.5/.69/.107), the **§11.4.169 test types** that gate it, and an **estimated complexity** (XS/S/M/L T-shirt, sizing-only TARGET — never a date, §11.4.6). It is **spec-only**: it describes the subtasks and how "done" is proven; it does not build them. Every subtask traces to its parent task's Desc/Deliverable/Acceptance in 07-...; nothing is invented beyond decomposing the stated work. These .k rows feed the workable-items DB (workable-items-model.md §7 docs\_chain context). Companion: dependency-graph.md (the DAG these subtasks inherit), subtask-deepening-p2.md, subtask-deepening-p3.md.

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## 0. Deepening conventions

Each parent task gets a subtask table: `id` (HVPN-P1-NNN.k, monotonic per parent, §11.4.54) · **Subtask** (≥6-word title, §11.4.91) · **Acceptance** (falsifiable, captured-evidence) · **Tests** (§11.4.169 codes — UNIT INT E2E FA SEC CHAOS STRESS PERF BENCH SCALE UI UX REC CHAL, the 07-... §1 vocabulary) · **Cx** (complexity XS≈0.5d, S≈1–2d, M≈2–3d, L≈3–5d — sizing TARGET).

A subtask is complete only when its test types are green with a `test_diary` evidence path (`workable-items-model.md` §9). Subtask complexity sums to ≈ the parent task's 07-... Effort; it is a sizing aid, not a commitment (§11.4.6, 07-... §22). Phase-0 entry-gate items (HVPN-P1-001..006) are prerequisites tracked in 06-...; they are not re-deepened here.

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## 1. E01 — Foundation (repo, proto, codegen)

**HVPN-P1-010 — Monorepo + submodule layout** (07-... §6; M)

| id | Subtask                                                                                                          | Acceptance                                                                       | Tests   | Cx |
|----|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------|----|
| .1 | Scaffold the snake_case component tree (helix-core/ helix-edge/ helix-go/ helix-proto/ helix-ui/ shims/ deploy/) | each dir present + decoupled (§11.4.28/.29); tree capture matches [SYNTHESIS §6] | UNIT,FA | S  |
| .2 |                                                                                                                  |                                                                                  | UNIT    | XS |

| id | Subtask                                                                                                              | Acceptance                                                                                                    | Tests  | Cx |
|----|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------|----|
|    | Wire <code>.gitmodules</code> + per-repo <code>helix-deps.yaml</code> (§11.4.31) for the six incorporated submodules | git submodule<br>status lists<br>containers/helix_qa/<br>challenges/<br>docs_chain/security/<br>vision_engine |        |    |
| .3 | upstreams/ recipes +<br>install_upstreams per repo<br>(§11.4.36)                                                     | git remote -v  <br>grep -c push =<br>expected count on a<br>fresh clone                                       | FA     | XS |
| .4 | Assert <b>no active CI</b><br>(§11.4.156): workflows<br>disabled/renamed                                             | CM-NO-ACTIVE-CI<br>green; git ls-files<br>finds no .github/<br>workflows/*.yaml                               | FA,SEC | XS |

#### HVPN-P1-011 — helix-`proto agent protobuf` + `buf codegen` (07–... §6; M)

| id | Subtask                                                                                                                        | Acceptance                                                       | Tests   | Cx |
|----|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------|----|
| .1 | Author <code>agent.proto</code><br>(Coordinator: Enroll/<br>WatchNetworkMap/<br>AdvertisePrefixes/ReportStatus)<br>per [02 §4] | buf lint clean;<br>service matches<br>the [02 §4]<br>contract    | UNIT    | S  |
| .2 | Wire buf generate → Go + Dart<br>+ Rust stubs                                                                                  | one source →<br>three generated<br>clients compile               | UNIT,FA | S  |
| .3 | buf breaking gate on<br>incompatible change                                                                                    | a removed field<br>FAILs buf<br>breaking<br>(captured)           | FA      | XS |
| .4 | Assert no hand-written client<br>code exists (§8 [04_P1])                                                                      | grep finds zero<br>hand-rolled RPC<br>client; generated-<br>only | SEC     | XS |

#### HVPN-P1-012 — REST OpenAPI + Dart/TS client gen (07–... §6; S)

| id | Subtask                                                                                                             | Acceptance            | Tests | Cx |
|----|---------------------------------------------------------------------------------------------------------------------|-----------------------|-------|----|
| .1 | Define <code>openapi/</code><br><code>helix.v1.yaml</code> (enroll-<br>tokens/devices/policies/<br>networks/stream) | OpenAPI 3.1 validates | UNIT  | S  |

| id | Subtask                                              | Acceptance                                                           | Tests   | Cx |
|----|------------------------------------------------------|----------------------------------------------------------------------|---------|----|
| .2 | Generate Dart + TS clients; round-trip a stub server | generated client round-trips; spec handler drift FAILs the gen check | UNIT,FA | S  |

#### HVPN-P1-013 — Local dev infra via containers submodule (07–... §6; S)

| id | Subtask                                                                                                  | Acceptance                                                  | Tests  | Cx |
|----|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------|----|
| .1 | internal/testinfra wrapper booting PG+Redis via pkg/boot/pkg/compose/pkg/health (§11.4.76/.161 rootless) | integration tests boot infra on demand; no ad-hoc docker    | INT,FA | S  |
| .2 | make test-infra-up/down with orphan-free teardown (§11.4.14)                                             | teardown leaves zero orphan containers (captured podman ps) | INT,FA | XS |

## 2. E02 — Store, RLS, no-log lint

*Risk-first (§11.4.132): the irreversible correctness floor — ships before any feature reads/writes data.*

#### HVPN-P1-020 — Migrations, enums, DB roles (07–... §7; M)

| id | Subtask                                                           | Acceptance                                                    | Tests    | Cx |
|----|-------------------------------------------------------------------|---------------------------------------------------------------|----------|----|
| .1 | goose DDL: 11 tenant tables + enums per [02 §2.2]                 | migrations apply+rollback clean on fresh PG; \dt matches spec | UNIT,INT | M  |
| .2 | Three roles helix_owner/helix_app(non-super,non-bypass)/helix_sys | role grants verified; helix_app lacks rolsuper/rolbypassrls   | INT,SEC  | S  |
| .3 | Assert no connection/traffic table present (privacy floor)        | \dt shows zero connections flows traffic packets table        | SEC      | XS |

#### HVPN-P1-021 — sqlc typed queries + store.WithTenant (07–... §7; M)

| id | Subtask                                     | Acceptance | Tests | Cx |
|----|---------------------------------------------|------------|-------|----|
| .1 | sqlc-generated compile-time-checked queries |            | UNIT  | S  |

| id | Subtask                                                                      | Acceptance                                                                                       | Tests    | Cx |
|----|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------|----|
|    |                                                                              | generated Queries<br>compiles; sqlc vet<br>clean                                                 |          |    |
| .2 | WithTenant(ctx,tid,fn) tx-<br>scoped SET LOCAL<br>app.tenant_id + WithSystem | a query forgetting<br>WHERE tenant_id<br>still returns only<br>the active tenant's<br>rows (INT) | UNIT,INT | M  |

#### HVPN-P1-022 — FORCE RLS + tenant\_isolation policies (07–... §7; M)

| id | Subtask                                                                                | Acceptance                                                                 | Tests    | Cx |
|----|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------|----|
| .1 | ENABLE+FORCE RLS +<br>tenant_isolation USING/<br>WITH CHECK on all 10 tenant<br>tables | every tenant table<br>has the policy; \d+<br>capture                       | UNIT,INT | M  |
| .2 | store/rls_guard.go startup<br>abort if helix_app can bypass<br>RLS                     | startup aborts under<br>a super role<br>(captured exit)                    | UNIT,SEC | S  |
| .3 | Cross-tenant denial proof                                                              | a crafted tenant-<br>A→tenant-B read<br>returns ZERO rows<br>(captured)    | SEC      | S  |
| .4 | Mid-tx drop rollback (no<br>leak)                                                      | connection drop<br>mid-tx rolls back;<br>no partial cross-<br>tenant write | CHAOS    | S  |

#### HVPN-P1-023 — No-logging CI schema-lint + mutation (07–... §7; S)

| id | Subtask                                                                       | Acceptance                                              | Tests    | Cx |
|----|-------------------------------------------------------------------------------|---------------------------------------------------------|----------|----|
| .1 | tools/schemalint rejecting<br>connection-log-shaped tables [02<br>§2.4]       | green on the<br>real schema;<br>pre-build gate<br>wired | UNIT,SEC | S  |
| .2 | Paired §1.1 mutation: inject<br>connections(src,dst,bytes,ts)<br>→ lint FAILs | the planted<br>table FAILs<br>the lint<br>(captured)    | SEC      | XS |

### 3. E03 — Identity, enrollment, PKI

#### HVPN-P1-030 — Dual identity: OIDC + anonymous device tokens (07-... §8; M)

| id | Subtask                                                      | Acceptance                                                                | Tests      | Cx |
|----|--------------------------------------------------------------|---------------------------------------------------------------------------|------------|----|
| .1 | OIDC managed mode (Keycloak/Authentik) → tenant/role mapping | an OIDC user maps to a tenant/role (INT)                                  | UNIT,INT   | M  |
| .2 | Anonymous device enroll-tokens (no PII, Mullvad posture)     | an anonymous device enrolls with zero PII persisted (SEC asserts DB/logs) | UNIT,SEC   | S  |
| .3 | Token mint/store-hashed/consume-atomic                       | single-use, short-lived; concurrent double-consume rejected (SEC)         | SEC,STRESS | S  |

#### HVPN-P1-031 — Enroll RPC (device key never leaves) (07-... §8; M · SLO2)

| id | Subtask                                                                             | Acceptance                                                           | Tests        | Cx |
|----|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------|----|
| .1 | Token verify → IPAM-allocate → issue cert → persist device row (public WG key only) | device row holds only the WG public key (SEC asserts no private key) | UNIT,INT,SEC | M  |
| .2 | Emit device.enrolled; first NetworkMap path                                         | event well-formed; map issued                                        | INT,E2E      | S  |
| .3 | SLO2 timing: enroll → first NetworkMap < 2 s                                        | captured histogram p99 < 2 s                                         | E2E,PERF     | S  |

#### HVPN-P1-032 — PKI: device-cert issuance + tenant CA (07-... §8; M)

| id | Subtask                                                                | Acceptance                                        | Tests    | Cx |
|----|------------------------------------------------------------------------|---------------------------------------------------|----------|----|
| .1 | pki.Issue/Rotate/Revoke short-lived (24 h) mTLS certs vs per-tenant CA | a cert authenticates WatchNetworkMap (INT)        | UNIT,INT | M  |
| .2 | CA-root secrecy (KMS/offline) + secret-leak audit (§11.4.10.A)         | CA private key never in any app log (audit green) | SEC      | S  |

**HVPN-P1-033 — Cert rotation + revoke-<1 s (07-... §8; M · SLO3, AC6)**

| id | Subtask                                                                   | Acceptance                                   | Tests    | Cx |
|----|---------------------------------------------------------------------------|----------------------------------------------|----------|----|
| .1 | Auto-rotate before expiry over the existing channel                       | rotation at the boundary works (captured)    | UNIT,INT | S  |
| .2 | device.revoke → drop WG peer from every map + edge enforce + cert revoked | revoked cert rejected immediately (SEC)      | SEC      | M  |
| .3 | SLO3 timing: revoke → edge enforcement < 1 s                              | captured revoke→enforcement timing p99 < 1 s | PERF     | S  |

**4. E04 — IPAM****HVPN-P1-040 — ULA /48 provisioning + AllocOverlayIP (07-... §9; M, D4)**

| id | Subtask                                                                     | Acceptance                                                       | Tests       | Cx |
|----|-----------------------------------------------------------------------------|------------------------------------------------------------------|-------------|----|
| .1 | Provision fd7a:helix:<rand>::/48 per tenant (: :1=gateway) on tenant create | each tenant gets a distinct ULA / 48 (INT)                       | UNIT,INT    | S  |
| .2 | AllocOverlayIP deterministic + concurrency-safe                             | 1000 concurrent allocations: no duplicate/gap (STRESS, captured) | UNIT,STRESS | M  |
| .3 | Restart-stable re-hydration of next_host                                    | allocation resumes after restart with no reuse                   | INT         | S  |

**HVPN-P1-041 — 4via6 for colliding IPv4 LANs (07-... §9; M)**

| id | Subtask                                     | Acceptance                                    | Tests    | Cx |
|----|---------------------------------------------|-----------------------------------------------|----------|----|
| .1 | Stable site-id allocator per connector site | each site gets a stable id (INT)              | UNIT,INT | S  |
| .2 | 4via6 route synthesis → Peer.allowed_ips    | one client reaches the same IPv4 CIDR via two | E2E      | M  |

| id | Subtask | Acceptance                                        | Tests | Cx |
|----|---------|---------------------------------------------------|-------|----|
|    |         | connectors, disambiguated by site (E2E curl each) |       |    |

## 5. E05 — Event backbone

**HVPN-P1-050 — `events.Bus` Redis Streams + `envelope`** (07-... §10; M, D3)

| id | Subtask                                                                                                          | Acceptance                                                         | Tests    | Cx |
|----|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------|----|
| .1 | <code>events.Bus</code> interface + Redis Streams <code>XADD/XReadGroup/XACK</code> + §5.2 <code>envelope</code> | at-least-once delivery; idempotent reaction; replay harmless (INT) | UNIT,INT | M  |
| .2 | Consumer-crash re-delivery on restart                                                                            | crash mid-handle re-delivers (CHAOS, captured)                     | CHAOS    | S  |

**HVPN-P1-051 — DLQ sweeper (`XAUTOCLAIM`) + `metric`** (07-... §10; S)

| id | Subtask                                                                                     | Acceptance                                                                | Tests          | Cx |
|----|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------|----|
| .1 | <code>XAUTOCLAIM</code> reclaim + dead-letter routing + <code>helix_events_dlq_total</code> | poisoned event lands in DLQ after N retries; metric increments (captured) | UNIT,INT,CHAOS | S  |

**HVPN-P1-052 — Event taxonomy producers** (07-... §10; S)

| id | Subtask                                                                                                                                                                                   | Acceptance                                                              | Tests    | Cx |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------|----|
| .1 | Typed constructors for the [04_P1 §5.3] taxonomy ( <code>device.</code> , <code>connector.</code> , <code>route.conflict</code> , <code>policy.*</code> , <code>gateway.failover</code> ) | each producer emits a well-formed envelope on its trigger (INT capture) | UNIT,INT | S  |

## 6. E06 — Policy model & compiler

**HVPN-P1-060 — Spec parse + group/host resolution** (07-... §11; M)



| id | Subtask                                                                 | Acceptance                                                                 | Tests    | Cx |
|----|-------------------------------------------------------------------------|----------------------------------------------------------------------------|----------|----|
| .1 | Parse policies.spec<br>jsonb (groups/hosts/<br>acls/exitNodes)          | valid spec parses;<br>malformed rejected                                   | UNIT     | S  |
| .2 | Resolve<br>group:*→device-ids +<br>host→CIDRs vs<br>advertised_prefixes | unknown group/host<br>rejected; ambiguity emits<br>route.conflict.detected | UNIT,INT | M  |
| .3 | Reject an ACL<br>granting a revoked<br>device                           | revoked-device grant<br>rejected (SEC)                                     | SEC      | S  |

**HVPN-P1-061 — Two-artifact compiler (AllowedIPs + verdict map)**  
(07–... §11; L · AC2/AC3)

| id | Subtask                                                                  | Acceptance                                                                           | Tests    |
|----|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|
| .1 | Pure default-deny<br>compile(tenant,spec)→{visible,allowedIPs,exitNodes} | golden-table<br>tests pass;<br>deterministic                                         | UNIT     |
| .2 | Port-level verdict map (nftables/eBPF granularity)                       | a contractor<br>granted<br>554,80<br>cannot reach<br>port 22<br>(verdict-map<br>SEC) | UNIT,SEC |
| .3 | Determinism property: same spec ⇒ byte-identical output                  | property test<br>green over N<br>runs<br>(§11.4.50)                                  | UNIT     |
| .4 | Perf: compile 1k-device tenant < 100 ms                                  | captured<br>timing < 100<br>ms                                                       | PERF     |

**HVPN-P1-062 — Dry-run fail-closed validation + atomic activate/  
rollback** (07–... §11; M · AC5)

| id | Subtask                                                                   | Acceptance                                                                     | Tests    | Cx |
|----|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------|----|
| .1 | Dry-run compile rejects<br>unknown ref / uncovered<br>dst / revoked grant | a bad policy never<br>reaches the active<br>version (fail-closed,<br>captured) | UNIT,SEC | M  |
| .2 |                                                                           |                                                                                | INT      | S  |

| id | Subtask                                                             | Acceptance                                                      | Tests | Cx |
|----|---------------------------------------------------------------------|-----------------------------------------------------------------|-------|----|
|    | Atomic version bump +<br>policy.compiled emit +<br>instant rollback | rollback restores the<br>prior compiled set<br>atomically (INT) |       |    |

## 7. E07 — Coordinator & WatchNetworkMap

*Critical path; the < 1 s convergence promise lives here.*

### HVPN-P1-070 — In-memory per-tenant topology graph (07–... §12; M)

| id | Subtask                                                                           | Acceptance                                                       | Tests     | Cx |
|----|-----------------------------------------------------------------------------------|------------------------------------------------------------------|-----------|----|
| .1 | Hydrate per-tenant<br>graph from PG on boot<br>(no durable coordinator<br>tables) | graph reconstructable<br>from PG + events alone<br>(INT)         | UNIT,INT  | M  |
| .2 | Event-driven<br>incremental subgraph<br>mutation                                  | applying an event<br>mutates only the<br>affected subgraph (INT) | INT,SCALE | M  |

### HVPN-P1-071 — buildMap(node) per-node map (07–... §12; M · AC2/AC3)

| id | Subtask                                                                                  | Acceptance                                                                                            | Tests    | Cx |
|----|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------|----|
| .1 | buildMap →<br>NetworkMap{self,gateway,peers,dns,transport}<br>with PeersVisibleTo filter | a node<br>never<br>appears in<br>another's<br>map unless<br>policy<br>grants it<br>(SEC,<br>captured) | UNIT,SEC | M  |
| .2 | relayEndpoint via gateway (MVP hub-and-spoke)                                            | reachability<br>matches<br>the Phase-0<br>slice                                                       | UNIT     | S  |

### HVPN-P1-072 — WatchNetworkMap stream loop (07–... §12; L · AC5)

| id | Subtask                          | Acceptance                        | Tests        | Cx |
|----|----------------------------------|-----------------------------------|--------------|----|
| .1 | mTLS-auth → snapshot/<br>resume- | reconnect with<br>known_version=N | UNIT,INT,E2E | L  |

| id | Subtask                                            | Acceptance                                                 | Tests | Cx |
|----|----------------------------------------------------|------------------------------------------------------------|-------|----|
|    | from-known_version → delta+keepalive loop          | resumes from deltas, no full resync (captured)             |       |    |
| .2 | Bounded per-stream send queue (slow-consumer shed) | a stalled consumer is shed without blocking others (CHAOS) | CHAOS | M  |

#### HVPN-P1-073 — Minimal-affected-set fan-out + reconcile metric (07-... §12; M · AC5/SLO1)

| id | Subtask                                                                      | Acceptance                                                                | Tests    | Cx |
|----|------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------|----|
| .1 | Compute minimal affected open-stream set; per-stream deltas (no full resync) | a 1-device change touches only the affected streams (INT count assertion) | UNIT,INT | M  |
| .2 | helix_reconcile_seconds histogram: event → delta-on-wire                     | SLO1 p99 < 1 s (captured histogram)                                       | PERF     | S  |

#### HVPN-P1-074 — 24 h soak / 10k-stream memory bound (07-... §12; M · SLO4)

| id | Subtask                                                      | Acceptance                                                   | Tests        | Cx |
|----|--------------------------------------------------------------|--------------------------------------------------------------|--------------|----|
| .1 | 10k simulated agents holding streams while flapping policies | SLO1 maintained under load (captured)                        | SCALE,STRESS | M  |
| .2 | 24 h RSS time-series, bounded no-leak                        | coordinator memory bounded, no growth over 24 h (SLO4 graph) | PERF,SCALE   | M  |

## 8. E08 — API surface + authz

#### HVPN-P1-080 — Connect handlers on Gin + mTLS resolver (07-... §13; M)

| id | Subtask | Acceptance | Tests    | Cx |
|----|---------|------------|----------|----|
| .1 |         |            | UNIT,INT | M  |

| id | Subtask                                                     | Acceptance                                            | Tests | Cx |
|----|-------------------------------------------------------------|-------------------------------------------------------|-------|----|
|    | One multiplexed server: Connect (HTTP/2) alongside Gin REST | native agent + gRPC-Web caller both reach Coordinator |       |    |
| .2 | authDevice(ctx) resolves identity from mTLS cert            | an unauthenticated agent RPC is rejected (SEC)        | SEC   | S  |

#### HVPN-P1-081 — REST routes + OpenAPI + RBAC (07–... §13; M)

| id | Subtask                                                    | Acceptance                                               | Tests    | Cx |
|----|------------------------------------------------------------|----------------------------------------------------------|----------|----|
| .1 | Gin routes (enroll-tokens/devices/policies/networks/audit) | routes match OpenAPI (INT)                               | UNIT,INT | M  |
| .2 | OIDC/session/API-token + RBAC (admin/operator/member)      | a member cannot mint enroll-tokens (403, SEC)            | SEC      | S  |
| .3 | RLS backstop (defense-in-depth)                            | RLS blocks cross-tenant read even if RBAC bypassed (INT) | INT,SEC  | S  |

#### HVPN-P1-082 — GET /v1/stream WS/SSE fan-out (07–... §13; S)

| id | Subtask                                            | Acceptance                                                                                             | Tests        | Cx |
|----|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------|----|
| .1 | Stream control-plane events to Console over WS/SSE | a policy apply surfaces "policy v N active" to a connected Console within the convergence window (E2E) | UNIT,INT,E2E | S  |

#### HVPN-P1-083 — AdvertisePrefixes + ReportStatus RPCs (07–... §13; S)

| id | Subtask                                                               | Acceptance                                                 | Tests    | Cx |
|----|-----------------------------------------------------------------------|------------------------------------------------------------|----------|----|
| .1 | AdvertisePrefixes accepts connector CIDs → connector.prefixes.changed | advertised prefixes flow into the coordinator graph (INT)  | UNIT,INT | S  |
| .2 | ReportStatus carries presence/transport/rtt only (no bytes/flows)     | SEC asserts the message has no byte/flow/destination field | SEC      | S  |

## 9. E09 — helix-core (Rust client/connector)

*The shared client/connector/edge crate; durable evolution of the Phase-0 interfaces.*

**HVPN-P1-090 — Transport trait + plain-UDP + MASQUE** (07–... §14; L · AC4, D1)

| id | Subtask                                                                                    | Acceptance                                                   | Tests        | Cx |
|----|--------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------|----|
| .1 | Harden the trait (send/recv/kind/effective_mtu, unreliable datagrams) into helix-transport | same trait drives client/connector/edge byte-for-byte        | UNIT         | M  |
| .2 | Production plain_udp impl                                                                  | throughput baseline captured (BENCH)                         | UNIT,BENCH   | S  |
| .3 | Production masque (quinn+h3+hand-rolled CONNECT-UDP/HTTP-Datagram, RFC 9298/9297/9221)     | MASQUE flow classifies HTTP/3, no WG signature (tshark, SEC) | UNIT,SEC,E2E | L  |
| .4 | MASQUE goodput vs plain ≥ 50%                                                              | captured BENCH ≥ 50% of plain                                | BENCH        | S  |

**HVPN-P1-091 — helix-wg boringtun wrapper** (07–... §14; M)

| id | Subtask                                          | Acceptance                                                      | Tests      | Cx |
|----|--------------------------------------------------|-----------------------------------------------------------------|------------|----|
| .1 | Wrap Tunn (handshake/encrypt/decrypt/4 verdicts) | in-process handshake + data round-trip identical plaintext      | UNIT       | M  |
| .2 | Sustained 1 GB transfer + 2-min rekey            | no handshake stall (STRESS); rekey at boundary works (captured) | E2E,STRESS | M  |

**HVPN-P1-092 — Orchestrator + network-map reconciler** (07–... §14; L · AC5)

| id | Subtask                                                             | Acceptance                          | Tests    | Cx |
|----|---------------------------------------------------------------------|-------------------------------------|----------|----|
| .1 | Three-loop orchestrator (tun→wg→transport, transport→wg→tun, timer) | loops drive a plain-UDP slice (E2E) | UNIT,E2E | M  |
| .2 | Reconciler consuming streamed MapUpdate                             | a delta adds/removes a peer with    | FA       | M  |

| id | Subtask                                                   | Acceptance                           | Tests | Cx |
|----|-----------------------------------------------------------|--------------------------------------|-------|----|
|    | (snapshot/delta) without restart                          | no reconnect of unrelated state (FA) |       |    |
| .3 | Mid-stream coordinator-restart re-sync from known_version | re-syncs cleanly (CHAOS)             | CHAOS | S  |

#### HVPN-P1-093 — Auto-escalation ladder (07–... §14; M · AC4)

| id | Subtask                                                                                              | Acceptance                                                                                  | Tests        | Cx |
|----|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------|----|
| .1 | Walk TransportPolicy.order (plain-udp→lwo→masque-h3) on handshake failure, honor allow_user_override | with plain WG blocked, client escalates to MASQUE and stays up (E2E transport=="masque-h3") | UNIT,E2E,SEC | M  |
| .2 | Deterministic ladder across N runs (§11.4.50)                                                        | identical escalation outcome over N=3                                                       | UNIT         | S  |

#### HVPN-P1-094 — Kill-switch + DNS-leak protection (07–... §14; L · AC7)

| id | Subtask                                                           | Acceptance                                                          | Tests    | Cx |
|----|-------------------------------------------------------------------|---------------------------------------------------------------------|----------|----|
| .1 | Core state machine drives platform firewall on Down/Reconnecting  | host-side pcap during a forced drop shows ZERO leaked packets (SEC) | UNIT,SEC | L  |
| .2 | DNS pinned to overlay resolver while up; no plaintext DNS on drop | zero plaintext DNS leaked (SEC capture)                             | SEC      | M  |
| .3 | State survives an abrupt interface flap                           | kill-switch holds across flap (CHAOS)                               | CHAOS    | S  |

#### HVPN-P1-095 — helix-ffi (flutter\_rust\_bridge v2) surface (07–... §14; M)

| id | Subtask                                                                 | Acceptance                                    | Tests      | Cx |
|----|-------------------------------------------------------------------------|-----------------------------------------------|------------|----|
| .1 | start(ClientConfig)/stop() + codegen                                    | Dart drives connect/disconnect; codegen clean | UNIT,FA    | S  |
| .2 | status_stream(StreamSink<TunnelStatus>) mirroring the broadcast channel | Dart renders live status from                 | UNIT,UI,FA | M  |

| id | Subtask | Acceptance                           | Tests | Cx |
|----|---------|--------------------------------------|-------|----|
|    |         | the stream alone (UI golden + FA ×3) |       |    |

## 10. E10 — helix-edge (data plane)

*Language fixed by Phase-0 G4 (default lean Rust, shares helix-transport).*

**HVPN-P1-100 — MASQUE termination on :443/udp + masquerade** (07-... §15; L · AC4)

| id | Subtask                                                      | Acceptance                                     | Tests    | Cx |
|----|--------------------------------------------------------------|------------------------------------------------|----------|----|
| .1 | Terminate MASQUE/H3, extract HTTP Datagrams → WG → kernel WG | a valid client tunnels through (E2E, captured) | UNIT,E2E | L  |
| .2 | Native decoy site for non-CONNECT-UDP probes                 | a scanner at :443 sees a plain website (SEC)   | SEC      | M  |
| .3 | CPU/Gbps within the G4 budget                                | captured BENCH within budget                   | BENCH    | S  |

**HVPN-P1-101 — Kernel-WG peer programming + relay** (07-... §15; M · AC2)

| id | Subtask                                          | Acceptance                                                                   | Tests    | Cx |
|----|--------------------------------------------------|------------------------------------------------------------------------------|----------|----|
| .1 | Program kernel WG peers from the coordinator map | peers appear in wg show (INT)                                                | UNIT,INT | M  |
| .2 | Hub-and-spoke relay peer↔peer via gateway        | client reaches an authorized connector LAN host through the relay (E2E curl) | E2E      | M  |

**HVPN-P1-102 — Edge verdict-map enforcement + revoke drop** (07-... §15; M · AC3/AC6, SLO3)

| id | Subtask                                                     | Acceptance                                          | Tests        | Cx |
|----|-------------------------------------------------------------|-----------------------------------------------------|--------------|----|
| .1 | Apply the compiler's port-level verdict map (nftables/eBPF) | an unauthorized port is dropped (AC3 SEC, captured) | UNIT,INT,SEC | M  |
| .2 |                                                             |                                                     | PERF,SEC     | S  |

| id | Subtask                                                          | Acceptance                                                                       | Tests | Cx |
|----|------------------------------------------------------------------|----------------------------------------------------------------------------------|-------|----|
|    | Drop a revoked device's sessions on device. <code>revoked</code> | revoke removes the kernel peer < 1 s (SLO3, captured); negligible p99 add (PERF) |       |    |

#### HVPN-P1-103 — Edge hardening posture (07–... §15; S)

| id | Subtask                                                            | Acceptance                                                       | Tests | Cx |
|----|--------------------------------------------------------------------|------------------------------------------------------------------|-------|----|
| .1 | Rootless Podman, read-only rootfs, seccomp, NET_ADMIN-only, no SSH | container runs unprivileged except NET_ADMIN (SEC)               | SEC   | S  |
| .2 | Fail-static: killing control plane doesn't drop data-plane tunnels | established tunnels survive a control-plane kill (CHAOS capture) | CHAOS | S  |

## 11. E11 — helix-ui (Flutter)

*All UI under OpenDesign tokens (§11.4.162) — light+dark, no ad-hoc CSS.*

#### HVPN-P1-110 — helix\_design system (OpenDesign tokens, Material 3) (07–... §16; M)

| id | Subtask                                                                                                                           | Acceptance                                                                                    | Tests   | Cx |
|----|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------|----|
| .1 | Brand-tokenized palette + signature components (ConnectButton/StatusChip/ExitPicker/ShieldIndicator/AdaptiveScaffold), light+dark | visual-regression goldens pass for every component in both themes (UI)                        | UNIT,UI | M  |
| .2 | No overlapping/overlaid labels (§11.4.162 layout invariant)                                                                       | layout-audit golden clean; palette reacts to connection state (REC vision-verified §11.4.159) | UI,REC  | S  |

#### HVPN-P1-111 — Shared shell `runHelixApp(flavor,...)` + Riverpod binding (07–... §16; M)

| id | Subtask                                             | Acceptance                                      | Tests   | Cx |
|----|-----------------------------------------------------|-------------------------------------------------|---------|----|
| .1 | One tree → three flavors (Access/Connector/Console) | same widget tree renders all three flavors (UI) | UNIT,UI | M  |



| id | Subtask                                                     | Acceptance                                                | Tests | Cx |
|----|-------------------------------------------------------------|-----------------------------------------------------------|-------|----|
| .2 | Riverpod providers = pure function of the FFI status stream | UI state deterministic over a replayed status stream (FA) | FA    | S  |

**HVPN-P1-112 — Access app screens (enroll → connect → reach) (07–... §16; L · AC9)**

| id | Subtask                                                                  | Acceptance                                                  | Tests      | Cx |
|----|--------------------------------------------------------------------------|-------------------------------------------------------------|------------|----|
| .1 | Enroll (paste/scan token) → connect → ExitPicker/authorized-network list | golden tests for each screen (UI)                           | UI         | M  |
| .2 | Real enroll→connect→reach-LAN journey window-scoped MP4 (§11.4.143/.159) | vision verdict confirms the authorized host loaded (UX/REC) | UX,E2E,REC | M  |

**HVPN-P1-113 — Console (web) admin screens (07–... §16; L · AC5/AC9)**

| id | Subtask                                                                                 | Acceptance                                                                                  | Tests      | Cx |
|----|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------|----|
| .1 | Devices/enroll-tokens/policy editor (validate→activate→rollback) + live /v1/stream feed | a policy edit reflects on devices < 1 s with a live "policy v N active" feed (E2E/REC, AC5) | UI,E2E,REC | L  |
| .2 | RBAC hides admin actions from member                                                    | member sees no admin action (SEC)                                                           | SEC        | S  |

**HVPN-P1-114 — Connector daemon UI/headless control (07–... §16; M · AC9)**

| id | Subtask                                                                     | Acceptance                                                                          | Tests      | Cx |
|----|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------|----|
| .1 | Enroll + advertise CIDRs + show prefixes/reachability; headless daemon mode | a connector advertises a LAN; its prefixes appear in Console + a client's map (E2E) | UI,E2E,REC | M  |

## 12. E12 — Platform tunnel shims

*MVP platforms only: iOS, Android, Linux. The shim is the only platform-specific code.*

### HVPN-P1-120 — Linux tun shim (Rust-native) (07-... §17; S · AC1/AC2)

| id | Subtask                                                          | Acceptance                                                                               | Tests  | Cx |
|----|------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------|----|
| .1 | Linux core drives the TUN directly (reference for the netns rig) | the Access Linux build runs the full enroll→connect→reach journey on the netns rig (E2E) | E2E,FA | S  |

### HVPN-P1-121 — Android VpnService + JNI shim (07-... §17; M · AC7/AC9)

| id | Subtask                                          | Acceptance                                         | Tests      | Cx |
|----|--------------------------------------------------|----------------------------------------------------|------------|----|
| .1 | VpnService owns the TUN; JNI loads helix-core.so | tunnel up on a real phone (device REC)             | E2E,UX,REC | M  |
| .2 | Kill-switch → VpnService always-on/ lockdown     | kill-switch blocks plaintext on drop (SEC capture) | SEC        | S  |

### HVPN-P1-122 — iOS NEPacketTunnelProvider shim (within G3 budget) (07-... §17; L · AC7/AC9)

| id | Subtask                                                                     | Acceptance                                     | Tests      | Cx |
|----|-----------------------------------------------------------------------------|------------------------------------------------|------------|----|
| .1 | NEPacketTunnelProvider configures settings + pumps packetFlow [?] Rust core | tunnel up on a real device (device REC)        | E2E,UX,REC | L  |
| .2 | Honor the G3 memory ceiling over a sustained transfer                       | captured Instruments RSS within ceiling (PERF) | PERF       | M  |
| .3 | Kill-switch + DNS-leak hold                                                 | zero plaintext/ DNS leak on drop (SEC)         | SEC        | S  |

## 13. E13 — Deploy

### HVPN-P1-130 — helixvpncctl bootstrap + ops CLI (Cobra) (07-... §18; M · AC1)

| id | Subtask                                                                | Acceptance                                                                  | Tests   | Cx |
|----|------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------|----|
| .1 | init generates tenant/admin/CA/overlay-/48/creds/gateway-keys/quadlets | helixvpncctl init on a fresh host produces a runnable deploy (FA, captured) | UNIT,FA | M  |
| .2 | gateway keys/enroll-token/policy apply/device revoke subcommands       | policy apply ./policy.jsonc validates + activates (E2E)                     | E2E     | S  |

#### HVPN-P1-131 — Podman quadlets (rootless, systemd-managed) (07-... §18; M · AC1)

| id | Subtask                                                                      | Acceptance                                                            | Tests  | Cx |
|----|------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------|----|
| .1 | Render helixd/helix-edge/postgres/redis quadlets, rootless, read-only rootfs | systemctl --user start helixvpn-pod brings the stack up rootless (FA) | INT,FA | M  |
| .2 | helixd connects to PG as the non-superuser role (RLS enforced)               | the app role cannot bypass RLS (SEC)                                  | SEC    | S  |

#### HVPN-P1-132 — On-demand integration infra via containers submodule (07-... §18; S)

| id | Subtask                                                                 | Acceptance                                                     | Tests  | Cx |
|----|-------------------------------------------------------------------------|----------------------------------------------------------------|--------|----|
| .1 | QA harness boots+tears down the full stack via the submodule (§11.4.76) | the E2E suite boots+tears down the stack itself (FA, captured) | INT,FA | S  |

#### HVPN-P1-133 — Compose dev-loop + K8s Phase-2 shape (reference) (07-... §18; S)

| id | Subtask                                                                        | Acceptance                                                                              | Tests | Cx |
|----|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|----|
| .1 | Docker Compose dev-loop (local) + reference stateless-coordinator K8s manifest | dev compose boots PG+Redis+helixd; the K8s manifest is lint-valid (not deployed at MVP) | INT   | S  |

## 14. E14 — QA, SLOs, DoD certification

*Anti-bluff capstone: the §11.4.40 full-suite retest, §11.4.107 recorded evidence, §11.4.27 100%-type coverage.*

### HVPN-P1-140 — netns + DPI + netem E2E rig (CI-ready) (07–... §19; M)

| id | Subtask                                                                                | Acceptance                              | Tests  | Cx |
|----|----------------------------------------------------------------------------------------|-----------------------------------------|--------|----|
| .1 | Promote the Phase-0 rig fed by the real control plane (lanA + nft WG-block + tc netem) | make e2e runs the full slice unattended | E2E,FA | M  |
| .2 | Re-runnable -count=3 identically (§11.4.98)                                            | three identical runs (captured)         | FA     | S  |

### HVPN-P1-141 — SLO instrumentation + dashboards (07–... §19; S · SLO1-4)

| id | Subtask                                                          | Acceptance                                                                                | Tests   | Cx |
|----|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------|----|
| .1 | Prometheus metrics + Grafana-as-code + alert rules at the §2 SLO | all four SLOs measured + asserted in CI thresholds (captured histograms, alert-on-breach) | PERF,FA | S  |

### HVPN-P1-142 — Security test suite (07–... §19; L · AC3/AC6/AC7/AC8)

| id | Subtask                                                                                                                          | Acceptance                                   | Tests     | Cx |
|----|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------|----|
| .1 | RLS cross-tenant denial + key-never-leaves + mTLS-reject + kill-switch/DNS pcap + revoke timing + secret-leak audit (§11.4.10.A) | every assertion green with captured evidence | SEC,SCALE | L  |
| .2 | Paired §1.1 mutations (drop an RLS policy / weaken kill-switch) FAIL their gate                                                  | each mutation FAILs (captured)               | SEC,CHAOS | M  |

### HVPN-P1-143 — Stress + chaos suite (§11.4.85) (07–... §19; M)

| id | Subtask                                                          | Acceptance                                        | Tests        | Cx |
|----|------------------------------------------------------------------|---------------------------------------------------|--------------|----|
| .1 | Sustained load + concurrency + boundary across the control plane | no deadlock/leak under 10× concurrency            | STRESS,SCALE | M  |
| .2 | Failure injection (Redis kill / PG                               | every fault recovers to a consistent state with a | CHAOS        | M  |

| id | Subtask                                        | Acceptance                     | Tests | Cx |
|----|------------------------------------------------|--------------------------------|-------|----|
|    | drop / coordinator<br>SIGKILL / disk-<br>full) | captured<br>recovery_trace.log |       |    |

#### HVPN-P1-144 — HelixQA Challenge bank + vision-verified recordings (07–... §19; M · AC1-AC9)

| id | Subtask                                                                                             | Acceptance                                                                                                        | Tests       | Cx |
|----|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------|----|
| .1 | challenges/helix_qa<br>bank dispatching real<br>user journeys, PASS<br>only on captured<br>evidence | the 8-AC journey set<br>scores PASS with a<br>vision-confirmed MP4<br>per AC (no frozen/bluff<br>frame §11.4.107) | CHAL,REC,UX | M  |
| .2 | Analyzer self-<br>validated (golden-<br>good/bad); broken<br>build FAILs                            | a deliberately broken<br>build scores FAIL                                                                        | CHAL        | S  |

#### HVPN-P1-145 — Full-suite retest + DoD certification gate (07–... §19; M · AC1-AC9/SLO1-4)

| id | Subtask                                                                                                                                                    | Acceptance                                                                                  | Tests       | Cx |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------|----|
| .1 | §11.4.40 release-gate<br>sweep on a clean baseline<br>(pre/post-build + on-<br>device + meta-mutation +<br>Challenge + no-log lint +<br>CONTINUATION sync) | all of §21 green<br>simultaneously on<br>a from-zero install<br>(§11.4.108 fresh<br>deploy) | FA,E2E,CHAL | M  |

## 15. E15 — Governance & release

#### HVPN-P1-150 — Workable-items DB projection + docs-chain sync (07–... §20; M)

| id | Subtask                                                                                               | Acceptance                                        | Tests       | Cx |
|----|-------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------|----|
| .1 | cmd/workable-items/<br>loader: parse leaves →<br>upsert items/test_diary<br>(git-tracked DB §11.4.95) | md↔db byte-<br>identical round-trip<br>(§11.4.93) | UNIT,INT,FA | M  |
| .2 |                                                                                                       | a leaf edit re-syncs<br>exports out-of-the-       | INT,FA      | S  |

| id | Subtask                                                     | Acceptance                    | Tests | Cx |
|----|-------------------------------------------------------------|-------------------------------|-------|----|
|    | .docs_chain/contexts/<br>wbs.yaml registered<br>(§11.4.106) | box; verify is the<br>CI gate |       |    |

**HVPN-P1-151 — Release tagging + multi-upstream publish** (07–... §20; S  
· AC1-AC9)

| id | Subtask                                                                                   | Acceptance                                                                                         | Tests | Cx |
|----|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------|----|
| .1 | Cut<br><HELIX_RELEASE_PREFIX>-1.0.0-mvp<br>(§11.4.151) on main + every owned<br>submodule | git tag -l<br>'<PREFIX>-*'<br>enumerates the<br>whole release<br>across repos                      | FA    | S  |
| .2 | Publish to all upstreams via merge-<br>onto-latest-main (no force-push<br>§11.4.113)      | the tag is<br>reachable on<br>every remote;<br>HVPN-P1-145<br>cert report is the<br>tag's evidence | FA    | S  |

## 16. Subtask roll-up + coverage note

| Epic                   | Parent tasks | Deepened subtasks | Notes                                         |
|------------------------|--------------|-------------------|-----------------------------------------------|
| E01 Foundation         | 4            | 12                | repo/proto/codegen/dev-infra                  |
| E02<br>Store+RLS+lint  | 4            | 11                | risk-first correctness floor                  |
| E03 Identity/<br>PKI   | 4            | 11                | revoke-<1s + key-never-<br>leaves             |
| E04 IPAM               | 2            | 5                 | ULA /48 + 4via6 (D4)                          |
| E05 Events             | 3            | 4                 | Redis Streams (D3)                            |
| E06 Policy<br>compiler | 3            | 9                 | two-artifact, default-deny                    |
| E07<br>Coordinator     | 5            | 11                | <1s convergence + 24h soak                    |
| E08 API+authz          | 4            | 8                 | Connect+REST+RBAC+RLS<br>backstop             |
| E09 helix-core         | 6            | 16                | Transport/WG/orchestrator/<br>kill-switch/FFI |

| Epic           | Parent tasks | Deepened subtasks | Notes                                   |
|----------------|--------------|-------------------|-----------------------------------------|
| E10 helix-edge | 4            | 9                 | MASQUE term + verdict-map + fail-static |
| E11 helix-ui   | 5            | 10                | OpenDesign + 3 flavors                  |
| E12 shims      | 3            | 6                 | Linux/Android/iOS                       |
| E13 deploy     | 4            | 7                 | helixvpnctl + quadlets                  |
| E14 QA/DoD     | 6            | 11                | rig/SLO/security/chaos/Challenge/cert   |
| E15 governance | 2            | 4                 | items-DB + release tag                  |
| <b>Total</b>   | <b>59</b>    | <b>134</b>        | —                                       |

Subtask complexity sums to  $\approx$  the parent task effort (07–... §22, ~273 person-days excl. Phase-0 prereqs); it is a **sizing aid, not a commitment** (§11.4.6). Every subtask is anti-bluff: its acceptance is a captured-evidence assertion (§11.4.5/.69/.107), metadata-only PASS is forbidden (§11.4.1), and it is complete only with a test\_diary evidence path (workable-items-model.md §9). The .k ids are monotonic per parent and never renumbered (§11.4.54). The full set of these subtasks feeds the §11.4.93 DB via the subtask-deepening-p1.md docs\_chain source (workable-items-model.md §7).

## Sources verified

- 07-phase1-mvp-wbs.md §6–§20 (every task Desc/Deliverable/Acceptance/Effort/Tests), §21 traceability, §22 effort roll-up — read 2026-06-26.
- 06-phase0-spike-wbs.md §0.1 (the 3-tier breakdown R5 aligns to) — read 2026-06-26.
- REFINEMENT\_NOTES.md R5 (subtask-tier asymmetry to close) — read 2026-06-26.
- Sibling workable-items-model.md (§3 field mapping, §6 test diary, §9 complete-definition), dependency-graph.md (the inherited DAG) — authored this volume.
- Constitution anchors  
§11.4.5/.6/.40/.54/.58/.69/.85/.91/.93/.98/.107/.108/.113/.132/.143/.151/.156/.159/.162/.169 — read 2026-06-26.

Honest boundary (§11.4.6): subtasks decompose the *stated* parent-task work; no new scope is invented. All complexity figures are sizing TARGETs, never date commitments; device-bound subtasks (iOS HVPN-P1-122., *Android HVPN-P1-121.*) carry the §11.4.3 SKIP-with-reason fallback where no device is reachable.